The ABJM $\mathfrak{su}(4)$ alternating chain: transfer matrices, ABJM point, localisation, discrete symmetries, selection rules and separation of variables

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Abstract

These notes collect everything obtained so far in the `su4-alternating` thread of the `abjm-c123` repository: the one-pass fused transfer matrices t_2, t_3; the identification of the ABJM point $\theta_A - \theta_F = 2h$ through the Minahan–Zarembo Hamiltonian; the localising transfer-matrix words and free-parameter rotations; the transfer-matrix construction of the ABJM three-point functions of the ABJM note; all highest-weight cyclic eigenstates and Q-functions up to $L = 4$ pairs; three discrete symmetries (LR, parity, and their product) in eigenvalue and Q-function form; vanishing rules for structure constants and length-independent parity selection rules; factorised bras annihilated by the symmetry differences; and the companion-twist separation-of-variables basis at four sites with sparse expansions of BPS states.

Contents

1 Overview, evidence levels, reproducibility **2**

2 Conventions and dictionaries **2**

3 Transfer matrices **3**

4 The ABJM point **3**

5 Localising words and free-parameter rotations (companion frame, generic θ) **4**

6 ABJM three-point functions from transfer matrices (diagonal frame) **4**

7 Highest-weight eigenstates and Q-functions **5**

8 Discrete symmetries **6**

9 Structure constants of the note's configuration **6**

10 Other three-point configurations; parity selection rules **6**

11 Factorised bras annihilated by the symmetry differences **7**

12 Separation of variables at four sites **7**

13 Open problems **8**

A Notebook and data index **9**

B Superseded or corrected statements **9**

1 Overview, evidence levels, reproducibility

Evidence levels. [exact]: rational/symbolic arithmetic in a wolfbook notebook. [num, 10^{-n}]: machine or high-precision numerics in a notebook, residual 10^{-n}. [numpy]: double-precision checks in the legacy `checks/*.py` scripts, not (yet) re-done in a notebook. All new work is done in wolfbook notebooks (`Paul/Mathematica/su4-alternating/*.wb`); the notebook containing each result is given in brackets.

Standard points. $h = \frac{1}{2}$. Generic: $\theta = (2/7, -3/5, 8/11, -11/8)$, companion twist $G_c(z)$ with $z = (3/2, 5/4, 2/3, 4/7)$. ABJM point: $\theta_A = \theta_F + 2h$ (homogeneous $\theta_A = 2h$, $\theta_F = 0$ for spectra; pair by pair $\theta_{A,n} = \theta_{F,n} + 2h$ with generic $\theta_{F,n}$ for SoV).

Data (in `data/`, Mathematica WXF, each with a `"description"` key documenting conventions): `abjm-hw-states.wxf` (all HW cyclic eigenstates, $L = 2, 3, 4$ pairs, with energies, LR data and C_ℓ), `abjm-tau.wxf` (eigenvalue polynomials $\bar{t}_{1,2,3}(u)$ of these states), `abjm-Qfunctions.wxf` (Q_1, Q_{12}, Q_{123}, roots, degrees), `abjm-tau-L4.wxf`. Load with `Import[file,"WXF"]`.

Headline results.

1. t_2 and t_3 are single-pass transfer matrices with explicit $\wedge^2$ and dual Lax operators (§3).
2. The ABJM chain is the point $\theta_A - \theta_F = 2h$; there $H_{\mathrm{MZ}} = U(0)^{-1}U'(0) + \text{const}$ with $U = \bar{t}_1\bar{t}_3$ (§4).
3. Three discrete symmetries $\mathcal{P}$ (LR), $\mathcal{Q}$ (parity), $\mathcal{PQ}$ with exact Q-function forms $Q_1 \propto Q_{123}(u+h)$, $Q_1 \propto Q_{123}(2h-u)$, $Q_1 \propto Q_1(h-u)$ (§8).
4. No LR selection rule for ABJM three-point functions in the two-block BPS class; length-independent *parity* selection rules exist, e.g. the note's configuration with a one-field bridge (§10).
5. The companion-twist SoV basis at four sites is a genuine B-eigenbasis at generic θ; BPS states that rotate only the F sublattice have ≤ 9 of 256 SoV coefficients; at the ABJM point the Ryan–Volin B is degenerate (§12).

2 Conventions and dictionaries

Chain. $2L$ sites ("L pairs"), A F A F …, periodic. A sites carry the $\mathbf{4}$: $\rho_A(E_{ij}) = e_{ij}$, group acts by g; F sites the $\bar{\mathbf{4}}$: $\rho_F(E_{ij}) = -e_{ji}$, group acts by g^{-T}. ABJM: $\mathrm{tr}(Y^{A_1}Y^\dagger_{B_1}\cdots)$; $L = 2, 3, 4$ pairs $= 4, 6, 8$ sites, dimensions $256, 4096, 65536$. $P_{ab} = \sum e_{ij} \otimes e_{ji}$ (same-type sites), $K_{ab} = \sum e_{ij} \otimes e_{ij}$ (A–F pair, $= |S\rangle\langle S|$, $S = \sum e_i \otimes e_i$). Bra site vectors transform as $v \to g^T v$ (A), $w \to g^{-1}w$ (F); the bilinear pairing $v \cdot w$ between an A and an F vector is invariant. (Paul's SoV naming calls A "anti-fundamental" and F "fundamental"; see the `README`. The $L = 2$-site notebook uses the opposite names.)

Lax, monodromy, fusion.

$$\mathcal{L}_s(u) = \sum_{ij} e_{ij} \otimes \big((u - \theta_s)\delta_{ij} + h\rho_s(E_{ij})\big), \qquad \mathcal{L}_A = u - \theta + hK, \quad \mathcal{L}_F = u - \theta - hP, \tag{1}$$

$T = \mathcal{L}_1\mathcal{L}_2\cdots\mathcal{L}_{2L}$ (site 1 leftmost), twist on the right, $T^G = TG$. Quantum minors: rows antisymmetrised, column k at $u - (k-1)h$; $t_a(u) = \sum_{|I|=a} \mathrm{qmin}_{I,I}[T^G](u)$, equivalently $t_a = \sum_{I,J} \mathrm{qmin}_{I,J}[T](u) \det G_{J,I}$.

Twists. Companion $G_c(z) = \{\{c_1, -c_2, c_3, -c_4\}, \{1,0,0,0\}, \{0,1,0,0\}, \{0,0,1,0\}\}$, $c_k = e_k(z)$; diagonal $\Lambda = \mathrm{diag}(z)$. Frame map [numpy] [`checks/chk_frame.py`]: with V the Vandermonde matrix (columns $(z_j^3, z_j^2, z_j, 1)^T$, $G_c V = V\Lambda$), $t^{G_c}(u)R = Rt^{\Lambda}(u)$, $R = \otimes(V^{-T}$ on A, V on F), $\langle x_{\mathrm{diag}}| = \langle x_{\mathrm{comp}}|R$. $\gamma := G^{-T}$.

Vacuum. $\langle 0| = e_1 \otimes e_4 \otimes e_1 \otimes e_4 \cdots$ (A: e_1, F: e_4). It is the Ryan–Volin SoV vacuum (§12). On the ket vacuum $T_{ij}|0\rangle = 0$ for $i < j$ and $T_{ii}|0\rangle = V_i|0\rangle$, $V_1 = (u - \theta_A + h)^L (u - \theta_F)^L$, $V_2 = V_3 = (u - \theta_A)^L (u - \theta_F)^L$, $V_4 = (u - \theta_A)^L (u - \theta_F - h)^L$ (pair-periodic θ).

Dictionary to Ryan–Volin (arXiv:2002.12341). RV Lax $u - \hbar \sum E_{ij} \otimes \pi(E_{ji})$, $T = \mathcal{L}_L \cdots \mathcal{L}_1$. With $\hbar = h$: $\pi = -\rho^T$, so our A site is RV's dual irrep and our F site RV's fundamental; our site s is RV's site $2L + 1 - s$. Quantum minors (2.15) and B (2.25) agree verbatim.

Dictionary to the ABJM note and 2603. Note (`StructuringABJM.pdf`): $O_2 = \mathrm{tr}(Y_1^\dagger Y^4)^L$, $O_3 = \mathrm{tr}(\tilde{Y}_1^\dagger \tilde{Y}^4)^L$, $\tilde{Y}_1^\dagger = Y_1^\dagger - Y_4^\dagger$, $\tilde{Y}^4 = Y^1 + Y^4$; (3.6) $C = \mathcal{N}(\langle Y_1^\dagger Y^4|^{L-M} \otimes \langle \tilde{Y}_1^\dagger \tilde{Y}^4|^M)|\Psi\rangle$. On one pair: $W_a := (e_1 + ae_4) \otimes (e_4 - ae_1) = \langle \mathrm{vac}|\rho(1 + \mathrm{aE}_{14})$; $W_0 = \mathrm{O}_2$ pair, $W_{-1} = \mathrm{O}_3$ pair. 2603 (arXiv:2603.26617): $\mathfrak{so}(6)$ chain of 6's, LR symmetry $\tau_+ = \tau_-$, structure constant $C = 0$ unless LR symmetric (Sec. IV, (18)–(20)); $\mathrm{O}_2 = \mathrm{O}_3$ with the first ℓ sites rotated by $1 - \kappa$ twist.

3 Transfer matrices

[`dual-lax-t3.wb`] From the one-site quantum minors (symbolic in u, θ, h) and checked identically against the quantum-minor t_a on the chain at 2 and 4 sites for companion, diagonal, generic and trivial twist [exact]:

$$t_3(u) = \prod_s g_s^{(3)}(u)\, \mathrm{tr}\big[\bar{\mathcal{L}}_1 \cdots \bar{\mathcal{L}}_{2L}\, \det G\, G^{-T}\big], \qquad \bar{\mathcal{L}}_A = (u - \theta + h) - hP, \quad \bar{\mathcal{L}}_F = (u - \theta - 3h) + hK,$$
$$g_A^{(3)} = (u - \theta - h)(u - \theta - 2h), \quad g_F^{(3)} = (u - \theta)(u - \theta - h), \tag{2}$$

$$t_2(u) = \prod_s g_s^{(2)}(u)\, \mathrm{tr}\big[\mathcal{L}_1^{(2)} \cdots \mathcal{L}_{2L}^{(2)}\, \wedge^2 G\big], \qquad g_A^{(2)} = u - \theta - h, \quad g_F^{(2)} = u - \theta,$$
$$\mathcal{L}_s^{(2)} = u - \tilde{\theta}_s + h \textstyle\sum_{ab} D(e_{ab}) \otimes \rho_s(E_{ab}), \tag{3}$$

with the sign gauge $(-1)^k$ on the dual index of t_3, D the derivation action on $\wedge^2\mathbb{C}^4$, $\tilde{\theta}_A = \theta$, $\tilde{\theta}_F = \theta + h$, $(\wedge^2 G)_{JI} = \det G_{J,I}$. Relative A/F shift: 0 (t_1), h (t_2), $4h$ (t_3); the earlier naive dual Lax missed the shift. $\bar{t}_a :=$ reduced transfer matrices (no $\prod g_s$). Vector form: one pass per vector (0.16 s for t_3 at 6 sites; a few seconds at 8 sites with the sparse site action), $[t_a, t_b] = 0$ to 10^{-16} at 6 sites.

4 The ABJM point

[`abjm-chain.wb`] Minahan–Zarembo two-loop Hamiltonian (arXiv:0806.3951; Bak–Rey arXiv:0807.2063; prefactor $\lambda^2/2$ dropped, form quoted from memory and confirmed by the commutation below):

$$H_{\mathrm{MZ}} = \sum_{l=1}^{2L} \big(2 - 2P_{l,l+2} + P_{l,l+2}K_{l,l+1} + K_{l,l+1}P_{l,l+2}\big). \tag{4}$$

Hermitian, $\mathfrak{su}(4)$-invariant, $[H_{\mathrm{MZ}}, S] = 0$ [exact]. $L = 2$ spectrum: 0 (×84), 4, 8 (×120), 12 (×30), 16 (×20), 20.

- [exact] ($L = 2$, u symbolic): $[H_{\mathrm{MZ}}, \bar{t}_a(u)] = 0$ for $a = 1, 2, 3$ at $\theta_A - \theta_F = 2h$, not at $\theta_A = \theta_F$; with $d = \theta_A - \theta_F$ symbolic the gcd of all commutator coefficients is $\propto d - 2h$: the point is unique. [num, 10^{-15}] at $L = 3$.

- $U(u) = \bar{t}_1(\theta_F + u)\bar{t}_3(\theta_A - h + u)$: at $u = 0$, $\bar{t}_1$ has P on F sites, $\bar{t}_3$ has P on A sites and the A–F dressings are $(x + hK)(-x - 4h + hK) = x(-x - 4h)$, $x = \theta_F - \theta_A$. Hence $U(0) = cS$, $c = h^4[x(-x - 4h)]^2$ at $L = 2$ [num, 10^{-16}], vanishing at $d \in \{0, 4h\}$. At the ABJM point $U(0) = S/16$ and $H_{\mathrm{MZ}} = U(0)^{-1}U'(0) + 12$ [exact].

Superseded: computations at $\theta_A = \theta_F$. There $U(0) = 0$, S drops out of the t-algebra, H_{MZ} does not commute with the t's, and an odd-momentum HW partner is degenerate with a cyclic state in t_1, t_2, t_3 (only S separates them); matching it through the untwisting limit required pair-periodic θ and a twisted shift $\hat{S} = S \circ \rho_{3..L}(\Lambda)$. All of this is an artefact of the singular point [`abjm-bridges.wb` §1–2].

5 Localising words and free-parameter rotations (companion frame, generic θ)

[`L2-ffbar-SoV-localisation.wb`, `L4-ffbar-free-rotations.wb`, `checks/`]

qKZ form. $\mathcal{L}_A(\theta) = hK$ (rank 1), $\mathcal{L}_F(\theta) = -hP$, so t_1 at an inhomogeneity transports the twist onto one site: at 2 sites $t_1(\theta_1) = h\rho_1(G^T)(\theta_{12} - hK)$, $t_1(\theta_2) = h\rho_2(G)(\theta_{12} - hK)$, $t_1(\theta_1)t_1(\theta_2)^{-1} = \rho(G^T)$ [exact]; at 4 sites $t_1(\theta_1) = hG_1^T R_{14}R_{13}R_{12}$, $t_1(\theta_2) = -hR_{23}R_{24}G_2R_{21}$, $t_1(\theta_3) = hR_{32}R_{31}G_3^T R_{34}$, $t_1(\theta_4) = -hG_4R_{41}R_{42}R_{43}$ with $R = \theta \pm hK$ (A–F) or $\theta \pm hP$ (same type) [exact].

Bare vacuum: every site localises [exact]. $X = t_1(\theta_1)^{-1} \to \rho_1(\gamma)$, $Y = t_1(\theta_2) \to \rho_2(\gamma)$ (const $h\theta_{12}\theta_{23}(\theta_{24} - h)$), $Z = t_1(\theta_3) \to \rho_3(\gamma^{-1})$ (const $h\theta_{32}(\theta_{31} + h)\theta_{34}$), $W = t_1(\theta_4)^{-1} \to \rho_4(\gamma^{-1})$; site vectors $e_1^T(1 - \kappa_1 G^{-T})$, $e_4^T(1 - \kappa_2 G)$, $e_1^T(1 - \kappa_3 G^T)$, $e_4^T(1 - \kappa_4 G^{-1})$. Overlaps $\prod(1 - \kappa_s \hat{o}_s)\langle 0|\Psi\rangle$ for single sites and adjacent pairs; same-type pairs pick up an exchange term, e.g. $\{1, 3\}$: $[(1-\kappa_1\hat{o}_1)(1-\kappa_3\hat{o}_3) - \frac{h}{\theta_{13}}\kappa_1\kappa_3(\hat{o}_1\hat{o}_3 - 1)]\langle 0|\Psi\rangle$; both ends $\{1, 4\}$ need longer words (singlet channel). Exchange dressing $\langle 0|t_1(\theta_1) = h\theta_{12}\theta_{14}[\theta_{13}\langle 0|\rho_1(G^T) + h\langle 0|\rho_3(G^T)]$. Global torus $\rho(X)$, $X \in \mathbb{C}[G_c^T]$, commutes with all t_a; κ's + torus: Jacobian rank 7. No fused T_λ ($|\lambda| \le 6$, 2288 tried) gives $G^{\pm 2}$ on a site: single words move a site only within $\mathrm{span}\{\mathrm{vac}, \mathrm{vac} \cdot \gamma^{\pm 1}\}$.

Generic globally rotated vacuum Ω: only the ends localise [exact]: $t_1(\theta_1)^{-1}$, $t_1(\theta_4)^{-1}$; end blocks $\langle\Omega|t_1(\theta_1)^{-1}t_1(\theta_2) = c_{12}\langle\Omega|\rho_1(G^{-T})\rho_2(G)$, $c_{12} = \theta_{23}(\theta_{24} - h)/((\theta_{13} + h)\theta_{14})$; of the 16 two-site SoV words only $\{1, t_3(\theta_1 - h)\} \times \{1, t_1(\theta_2)\}$ localise on block $\{1, 2\}$ (2603 (G2) analogue).

Common rotations and good references [numpy]. $\langle 0|(1 + a\,t_1(\theta_1) + b\,t_1(\theta_3) + c\,t_1(\theta_1)t_1(\theta_3)) = \langle 0|\rho_1(g)\rho_3(g)$, $g = 1 - \kappa G_c^T$, $a = -\kappa/(\alpha\theta_{13})$, $b = -\kappa(\theta_{13} - h)/(\theta_{13}l_3)$, $c = \kappa^2/m$; this is a *global* rotation $\exp(-\kappa' E_{12})$ (it fixes e_4). The $\{2, 4\}$ analogue is generated by E_{34}. A global rotation $v \otimes w \otimes v \otimes w$ ($v \cdot w = 0$) is as good a reference as $\langle 0|$ iff $wGv^T = 0$. With $v = e_1^T p(G^T)$, $w = e_4^T q(G)$: global iff $\deg p + \deg q \le 2$, good iff ≤ 1. The nine $\langle 0|X^a Y^b$ ($X = t_1(\theta_1)$, $Y = t_1(\theta_3)$, $a, b \le 2$) span $\{e_i \otimes e_4 \otimes e_j \otimes e_4 : i, j \le 3\}$ (contains $(G^T)^2$ on one site). On the 2-parameter state Ω_2 (not a good reference) only end-growing words localise, as fixed twist rotations.

6 ABJM three-point functions from transfer matrices (diagonal frame)

[`ABJM-3pt-transfer-matrices.md`, `abjm-bridges.wb`] [numpy] unless stated.

Identification. A pair twist rotation $\rho_{\mathrm{pair}}(D)$, $D = \mathrm{diag}(d)$, maps $W_a \to W_{ad_4/d_1}$; $D_\kappa = 1 - \kappa\Lambda$ gives $W_a \to W_{a(1-\kappa z_4)/(1-\kappa z_1)}$, and $\kappa = 1/z_4$ turns an O_3 pair into an O_2 pair. So (3.6) is $\langle W_{-1}^L|$ with its first pairs rotated by D_{1/z_4}, the analogue of 2603's O_2. The companion SoV vacuum is a

global rotation with all components nonzero in the diagonal frame, so the 3-pt reference is taken directly as a diagonal-frame global rotation (as in 2603).

Pair localisation. On $\langle W_{-1}|^{\otimes 2}$, of the 16 words $t_a(\theta_1-h)t_1(\theta_2)^k$ exactly $\{1, t_3(\theta_1-h)\}\times\{1, t_1(\theta_2)\}$ localise on pair 1; all 320 pair-1-localising words (up to two library factors per site) span only 4 dimensions and give product states only $\rho_1(\Lambda^i)\rho_2(\Lambda^j)$ with $(i,j) \in \{(0,0),(-1,0),(-1,1)\}$. The even bridge needs Λ on the F site alone (fourth type), which is missing: *the F-site gap.* Global torus: $t_1(\theta_1)^{-1}t_1(\theta_2)t_1(\theta_3)^{-1}t_1(\theta_4) = c\,\rho(\gamma)$.

Odd bridge (one field): exact at any twist.

$$\langle W|t_1(\theta_1)^{-1} = c_1\langle W|\rho_1(\Lambda^{-1}), \qquad 1/c_1 = h\prod_{F\,s}\theta_{1s}\prod_{A\,s\neq 1}(\theta_{1s}+h),$$

hence

$$\langle e_1\otimes(e_4+e_1)\otimes W_{-1}^{\otimes(L-1)}|\Psi\rangle = \frac{1-z_4/(c_1\tau_1(\theta_1))}{1-z_4/z_1}\,\langle W_{-1}^{\otimes L}|\Psi\rangle. \tag{5}$$

Verified through the untwisting limit at the ABJM point with a generic inhomogeneous deformation, 10^{-11}–10^{-14} [`abjm-bridges.wb` §3]; $\langle W|\Psi\rangle \sim \epsilon$, bracket $\sim 1/\epsilon$.

Highest-weight trick. For HW Ψ, $\langle \text{vac}\otimes W_{-1}|\Psi\rangle = \langle W_b\otimes W_{b-1}|\Psi\rangle$ for all b (10^{-16}); with $b = z_1/(z_1-z_4)$, $\langle W_{b-1}^{\otimes 2}|t_1(\theta_1)^{-1}t_1(\theta_2) = c\langle W_b\otimes W_{b-1}|$ exactly at finite twist, but $b\to\infty$ as $z\to 1$ and the untwisting limit is wrong and state dependent. 2603 avoids this with a continuous κ-family; for ABJM pairs that family is the open problem.

7 Highest-weight eigenstates and Q-functions

States [`abjm-LR-symmetry.wb`] [`num`, 10^{-13}]. All HW cyclic ($S=1$) joint eigenstates of $\bar t_1, \bar t_3$ at the ABJM point: $6, 26, 152$ at $L = 2, 3, 4$; all are H_{MZ} eigenstates ($\le 3.5\cdot 10^{-13}$); joint spectrum non-degenerate (gap ≥ 0.0065). (Pitfall found and fixed: degenerate kernel vectors returned by `Eigensystem` need not be orthonormal.)

Q-functions [`abjm-Qfunctions.wb`] [`num`, 10^{-11}]. On the vacuum $\sum_a(-1)^a\tau_a D^a = (1-V_1D)(1-V_2D)(1-V_3D)(1-V_4D)$, $D = e^{-h\partial_u}$ (checked numerically), so the polynomial gauge is $Q\cdot\Phi$, $\Phi(u) = V_4(u)\Phi(u-h)$:

$$\sum_{a=0}^{4}(-1)^a\tau_a(u)\,Q(u+h-ah)\prod_{b=a}^{3}V_4(u-bh) = 0, \tag{6}$$

$\tau_2 = \prod g^{(2)}\bar t_2$, $\tau_3 = \prod g^{(3)}\bar t_3$, $\tau_4 = \text{qdet}$ (A site: $(u-\theta+h)(u-\theta-h)(u-\theta-2h)(u-\theta-3h)$). Vacuum solutions have degrees $0, L+1, L+2$. Q_1, Q_2, Q_3 = polynomial solutions ordered by degree; $Q_{12} = \text{Cas}(Q_1,Q_2)/\text{Cas}_{\text{vac}}$, $Q_{123} = \text{Cas}(Q_1,Q_2,Q_3)/\text{Cas}_{\text{vac}}$ (shifts $0,-h,-2h$). For all 184 states $\deg(Q_1,Q_{12},Q_{123}) = (K_1,K_2,K_3)$,

$$K_2 = L - \tfrac{p+r}{2} - q, \qquad K_1 = \tfrac{L-r+K_2}{2}, \qquad K_3 = \tfrac{L-p+K_2}{2} \qquad (\text{Dynkin } [p,q,r]), \tag{7}$$

Casoratian remainders $\le 8\cdot 10^{-12}$. Eigenvalue polynomials by interpolation at $2L+1$ points, 8 parallel subkernels at $L=4$.

8 Discrete symmetries

[`abjm-LR-symmetry.wb`, `abjm-other-symmetries.wb`] [`num`, 10^{-16}]. Operators commuting with $H_{\rm MZ}$: $\mathcal{P}$ = (Weyl reversal $e_a \mapsto e_{5-a}$)$\circ$(one-site shift), $\mathcal{Q}$ = (signed Weyl $e_1 \to e_4$, $e_2 \to e_3$, $e_3 \to e_2$, $e_4 \to -e_1$)$\circ$(reflection $s \mapsto 2L+1-s$):

$$\mathcal{P}\bar{t}_1(u)\mathcal{P}^{-1} = \bar{t}_3(u+h), \quad \mathcal{Q}\bar{t}_1(u)\mathcal{Q}^{-1} = \bar{t}_1(2h-u), \quad \mathcal{P}^2 = S, \quad \mathcal{Q}S\mathcal{Q}^{-1} = S^{-1}. \tag{8}$$

$\mathcal{P}$ maps HW states of $[p,q,r]$ to HW states of $[r,q,p]$ (it exchanges the $\mathbf{4}$ and $\bar{\mathbf{4}}$ slots). Symmetric states (eigenvalue polynomials exact to 10^{-8} relative; Q-function forms checked with fixed shifts on all 184 states):

symmetry	eigenvalues	Q-functions	also
LR ($\mathcal{P}$)	$\tau_1(u) = \tau_3(u+h)$	$Q_1(u) \propto Q_{123}(u+h)$	
P ($\mathcal{Q}$)	$\tau_1(u) = \tau_1(2h-u)$	$Q_1(u) \propto Q_{123}(2h-u)$	$\tau_2(u) = \tau_2(c-u)$
LP ($\mathcal{PQ}$)	$\tau_1(u) = \tau_3(3h-u)$	$Q_1(u) \propto Q_1(h-u)$	$\tau_2(u) = \tau_2(c-u)$

Reflection turns the Baxter equation into its dual, hence P exchanges $Q_1 \leftrightarrow Q_{123}$ and LP maps Q_1 to itself. Every non-LR state has an LR partner $\mathcal{P}\Psi$ with $Q_1^{\rm partner} = Q_{123}(u+h)$.

L	states	LR	P	LP	all three	LR only	P only	LP only	none
2	6	4	6	4	4	0	2	0	0
3	26	8	18	12	8	0	10	4	4
4	152	24	60	40	20	4	40	20	68

$\mathcal{Q}$-eigenstates have eigenvalue $(-1)^p$ in this sign convention.

9 Structure constants of the note's configuration

[`abjm-bridges.wb`, `abjm-LR-symmetry.wb`] Bra for bridge ℓ: first ℓ sites O_2 fields (A e_1, F e_4), the rest O_3 fields (A $e_1 - e_4$, F $e_4 + e_1$); $C_\ell = \langle B_\ell|\Psi\rangle/\|\Psi\|$ (bilinear bra, phase of Ψ fixed by its largest component).

$L=2$ state	$E_{\rm MZ}$	nonzero C_ℓ
$[2,0,2]$ vacuum	0	all 1
$[0,2,0]$	16	—
$[1,0,1]$ parity pair	12, 12	$C_1 = C_3 = (1 \mp i\sqrt{3})/\sqrt{48}$, $\lvert C_1\rvert = 1/\sqrt{12}$
singlet	20	$C_2 = -1/\sqrt{40}$
singlet	4	$C_2 = 1/\sqrt{24}$

(Closed forms are numerical recognitions.) Vanishing at $L = 2,3,4$: (i) R-charge: unless $\Psi \in [p,0,p]$ and $L - p \le \ell \le L + p$; (ii) bridge parity, kinematic on the whole HW cyclic sector: if $L+p$ is odd, $C_\ell = 0$ unless $\ell \equiv L + p \pmod 2$; (iii) *no LR selection rule*: non-LR states have nonzero C_ℓ; (iv) isolated zeros at $L = 4$: the LR $[1,0,1]$ state at $E = 32$ (all ℓ) and the LR $[2,0,2]$ state at $E = 8$ (only even ℓ).

10 Other three-point configurations; parity selection rules

[`abjm-3pt-configs.wb`] Two-block BPS bras: block 1 on ℓ sites with (a_1, f_1), block 2 on the rest with (a_2, f_2), $a_i \cdot f_i = 0$, *same polarisations for every* L. Up to $GL(4)$ and site rescalings the only invariants are whether the cross pairings $a_1 \cdot f_2$, $a_2 \cdot f_1$ vanish (plus linear dependences), giving six classes; with block 1 = (e_1, e_4), block 2: generic $(e_1 - e_4, e_4 + e_1)$ (the note's), $a_2 \cdot f_1 = 0$: $(e_2, e_1 + e_3)$, $a_1 \cdot f_2 = 0$: $(e_4 + e_2, e_3)$, both zero: (e_2, e_3), $a_2 = a_1$: (e_1, e_3), $f_2 = f_1$: (e_2, e_4); bridge starting on A or F; whole multiplets probed by random global rotations; 180 configurations at $L = 2,3,4$.

- LR: no configuration couples only to LR states at $L = 3, 4$.
- P: length-independent selection rules ($C = 0$ unless $Q_1(u) \propto Q_{123}(2h-u)$), verified at operator level on the cyclic sector, $\langle B|X_P(u)|r\rangle = 0$ for cyclic r, 10^{-16} at $L = 3, 4$ (vacuous at $L = 2$):

block 2	bridge ℓ	start
generic (note's W_{-1})	1 or $2L-1$	A or F
$a_2 \cdot f_1 = 0$	1	A
$a_1 \cdot f_2 = 0$	1	F
both cross pairings zero	2 or $2L-2$	A or F

  Control: the note's configuration at $\ell = 2$ fails ($10^{-2.4}$). The full operator identity (without the cyclic restriction) fails, as it must.
- LP: no uniform rule.

11 Factorised bras annihilated by the symmetry differences

[`abjm-LR-factorized.wb`, `abjm-other-symmetries.wb`] $X_{\rm LR} = \bar{t}_1(u) - \bar{t}_3(u+h)$, $X_{\rm P} = \bar{t}_1(u) - \bar{t}_1(2h-u)$, $X_{\rm LP} = \bar{t}_1(u) - \bar{t}_3(3h-u)$.

- LR, $L = 2$, untwisted: left kernel 136/256 ($X_{\rm LR}$ linear in u); product states in it only the BPS orbit. Twisted, $\Lambda = \mathrm{diag}(z_1, z_2, 1/z_2, 1/z_1)$ ($z_1 z_4 = z_2 z_3 = 1$, so the t_3 twist $\det\Lambda\,\Lambda^{-1}$ is the Weyl image of Λ): 36 LR states, kernel 36, product states only $e_a \otimes e_{5-a}$; the note's W_a only at $a = 0$.
- P: every two-site state $(v \otimes w)^{\otimes L}$ is annihilated (10^{-16}, $L = 2, 3, 4$) – integrable two-site boundary states.
- LP: product solutions at $L = 2$ (SU(2)×SU(2)-sector states) fail at $L = 3, 4$.

(Per-length product searches are superseded by the length-independent requirement of §10.)

12 Separation of variables at four sites

[`sov-companion-L4.wb`] Four sites A F A F, companion twist $G_c(z)$.

Basis. Words

$$\langle a_1, k_2, a_3, k_4| = \langle 0|\, t_{a_1}(\theta_1 - h)\, t_1(\theta_2)^{k_2}\, t_{a_3}(\theta_3 - h)\, t_1(\theta_4)^{k_4}, \qquad a_i, k_i \in \{0, 1, 2, 3\}, \tag{9}$$

rank 256 [exact], overlaps $\langle\Lambda|\Psi\rangle = \tau_{a_1}(\theta_1 - h)\tau_1(\theta_2)^{k_2}\tau_{a_3}(\theta_3 - h)\tau_1(\theta_4)^{k_4}\langle 0|\Psi\rangle$.

SoV vacuum. RV's $\langle 0|$ (4.12) is the lowest-weight GT vacuum: (3.19) $\langle 0|T_{j1} = \delta_{j1}\nu\langle 0|$ applied recursively through the embeddings, i.e. killed by every strictly lower-triangular $T_{ij}(u)$ ($i > j$) of the untwisted monodromy and eigen of the T_{ii}. (3.19) alone leaves a 15-dim space; the full condition has a unique solution, equal to our $\langle 0|$ [exact] (generic and ABJM θ).

B-operator. RV (2.25): $B(u) = \sum_{k_1\in\{1,2,3\}} \sum_{K_2} T_{k_1,1}(u)\, \mathrm{qmin}_{K_2;(1,k_1+1)}(u+h)\, \mathrm{qmin}_{(1,2,3);(1,K_2+1)}(u+2h)$, untwisted T; $[B(u), B(v)] = 0$ [exact]. At generic θ all 256 words are B-eigenbras [exact]: a genuine SoV basis, equal to RV's up to normalisation.

BPS expansions (generic θ; number of nonzero coefficients of 256; unchanged at the shifted ABJM-point word bases):

BPS state $v \otimes w \otimes v \otimes w$, $v \cdot w = 0$	$\#\{c_\Lambda \neq 0\}$
vacuum	1
F only: $v = e_1$, $w \in e_1^\perp$ general	9 ($a_1 = a_3 = 0$, $k \leq 2$)
F only: $\exp(\kappa E_{34})$	4 ($k \leq 1$; not a product: F–F exchange term)
F only: $\exp(\kappa E_{24})$	5
A only: $\exp(\kappa E_{12})$	13 ($a \leq 2$, $k \leq 1$)
A only: $w = e_4$, $v \in e_4^\perp$ general	56
SU(2)×SU(2) sector	37
torus $\mathbb{C}[G_c^T]$	134–236
generic	256

$\exp(\kappa E_{34})$ is the $\{2,4\}$ common rotation and $\exp(\kappa E_{12})$ the $\{1,3\}$ one of §5.

Localisation on sites 1,2 [exact]. Of the 16 site-1,2 words exactly $\{1, t_3(\theta_1 - h)\} \times \{1, t_1(\theta_2)\}$ leave sites 3,4 at the vacuum (they give e_1e_4, e_1e_3, e_4e_4, e_4e_3 on sites 1,2), and the constants factorise:

$$\langle 0|\big(1 + \alpha t_3(\theta_1 - h)\big)\big(1 + \beta t_1(\theta_2)\big) = (e_1 + \alpha k_3 e_4) \otimes (e_4 + \beta k_1 e_3) \otimes e_1 \otimes e_4, \tag{10}$$

$k_1 = -2263/14000$, $k_3 = -21489/120736$ at the generic point. The localised span is 5-dim; the extra element $\langle 0|[t_1(\theta_2)^2 - \frac{15841}{1450} t_1(\theta_1 - h)t_1(\theta_2)] = (e_1 \otimes e_2 + \frac{325}{357} e_2 \otimes e_3) \otimes e_1 \otimes e_4$ is entangled (reaches $e_4^T G_c^2$ on site 2) and adds no product states.

Localisation on sites 1,2,3 [exact]. Localised single words $\{1, t_3(\theta_1 - h)\} \times \{1, t_1(\theta_2)\} \times \{1, t_3(\theta_3 - h)\}$, span 15-dim; $t_3(\theta_3 - h)$ on $\langle 0|$ gives an F_2–A_3 singlet plus an exchange term with site 1. Product states: (A) $(e_1 + ae_4) \otimes (e_4 + ce_3) \otimes e_1 \otimes e_4$; (B) $(e_1 + ae_4) \otimes e_3 \otimes (e_1 + be_4) \otimes e_4$ (arbitrary a, b; words $\langle 0|t_1(\theta_2) \times$ degree ≤ 1 in $t_3(\theta_1 - h)$, $t_3(\theta_3 - h)$, with an exchange correction). The three-site generic $(e_1 + ae_4) \otimes (e_4 + ce_3) \otimes (e_1 + be_4)$ is not localisable.

ABJM point. RV's genericity assumption (2.9) $\theta_\alpha - \theta_\beta \notin \hbar\mathbb{Z}$ fails. At $\theta_{A,n} = \theta_{F,n} + 2h$: the fused $t_3(\theta_A - h)$ vanishes (scalar factor), the reduced words have rank $169 = 13^2$ (same-pair F Lax $h(1 - P)$ at $u = \theta_F + h$); words at $u_A = \theta_A - 3h$ or $\theta_A + 2h$ are a basis but only the 16 F-only words are B-eigen; B itself has only 169 distinct eigenvalues for $d = \theta_A - \theta_F \in \{h, 2h, 3h\}$ (256 for $d \in \{0, 4h\}$ and generic) [num, 10^{-20}].

13 Open problems

1. SoV at the ABJM point: limit $d \to 2h$ of the generic SoV basis (singular measure), or a different B (e.g. from the $\bar{\mathbf{4}}$ auxiliary space, another frame) that stays simple at $d = 2h$.

2. Map the physical (diagonal-frame) 3-pt polarisations into the sparse companion-frame families; two-block bras from F-only blocks; closed forms of SoV coefficients.

3. Even bridges: the F-site-only twist rotation (κ-family on a pair); $L = 3$ pairs for the transfer-matrix construction; homogeneous limit of the odd-bridge formula.

4. Mechanism of the $\ell = 1$ (and both-zero $\ell = 2$) parity rules; $L = 5$; non-BPS two-block families; transfer-matrix bridges as in 2603.

5. Twisted versions (2603 works at generic twist with $\omega = 1$); LR-compatible non-diagonal twists; integrability of $\langle W_a^L|$ (note's (4.8)).

A Notebook and data index

file	content
`L2-ffbar-SoV-localisation.wb`	2 sites: qKZ, spectrum, B, SoV basis, one-site rotations
`L4-ffbar-free-rotations.wb`	4 sites: localisation inventory, κ-families, torus, fused-T scan, reach
`dual-lax-t3.wb`	one-pass t_3 (dual Lax) and t_2 ($\wedge^2$ Lax), checks, vector form
`abjm-chain.wb`	H_{MZ}, commutation iff $\theta_A - \theta_F = 2h$, H_{MZ} from U, $L = 3$ check
`abjm-bridges.wb`	bridge overlaps; $\theta = 0$ (superseded), shift test, ABJM-point table and odd-bridge formula
`abjm-LR-symmetry.wb`	$\mathcal{P}$, $\mathcal{Q}$; all HW cyclic states $L = 2, 3, 4$; C_ℓ vanishing rules; saves states
`abjm-Qfunctions.wb`	Baxter equation, Q_1, Q_{12}, Q_{123}, LR in Q-language; saves Q's and τ's
`abjm-3pt-configs.wb`	two-block configuration scans (LR, P, LP); parity selection rules
`abjm-LR-factorized.wb`	factorised LR-annihilated bras, untwisted and twisted ($L = 2$)
`abjm-other-symmetries.wb`	P and LP symmetries, Q-forms, factorised P/LP bras
`sov-companion-L4.wb`	SoV basis at 4 sites, B and vacuum checks, BPS expansions, localisation, ABJM point
`ABJM-3pt-transfer-matrices.md`	write-up of the transfer-matrix 3-pt construction (numpy era)
`data/*.wxf`	states, τ polynomials, Q-functions

B Superseded or corrected statements

- Bridge tables and degeneracy analyses at $\theta_A = \theta_F$ (singular point; replaced by the ABJM point).
- The twisted-shift fix $\hat{S}$ (artefact of $\theta_A = \theta_F$).
- "The SoV basis survives at the ABJM point with shifted evaluation points": the shifted words are a basis but not B-eigen.
- Per-length product-state searches for P/LP/LR (the 3-pt bras must be length independent).
- Odd-bridge formula: the factor $1/(1 - z_4/z_1)$ was missing in an early version (corrected 11-10-2026).

References. B. Bargheer, C. Bercini, B. Lefundes, P. Ryan, arXiv:2603.26617; J. A. Minahan, K. Zarembo, arXiv:0806.3951; D. Bak, S.-J. Rey, arXiv:0807.2063; P. Ryan, D. Volin, arXiv:1810.10996, arXiv:2002.12341; the ABJM note `StructuringABJM.pdf`.